CHAPTER 7

Ethernet

IEEE 802.3 frame · frame-size limits · efficiency · hubs vs switches · collision and broadcast domains

QUICK TRICK

Ethernet frame: **Dest 6 + Src 6 + Type 2 + Data 46–1500 + FCS 4 = 64 to 1518 bytes**. Preamble (7) + SFD (1) are extra. Hub = **one** collision domain. Switch = **one per port**. Router = **one broadcast domain per interface**.

7.1 What is Ethernet?

● LEVEL 1 · BEGINNER

Ethernet (IEEE 802.3) is the most common wired LAN. It frames data, uses 48-bit MAC addresses, and (in its half-duplex form) uses CSMA/CD from Chapter 6.

7.2 Frame format

● LEVEL 2 · EXAM

```
| Preamble 7 | SFD 1 | Dest MAC 6 | Src MAC 6 | Type/Len 2 | Data 46..1500 | FCS 4 |
|<-- sync, not counted -->|<------------ frame size 64 .. 1518 bytes ------------>|
```

Field	Size	Purpose
Preamble + SFD	7 B + 1 B	Receiver clock synchronisation (10101010... then 10101011)
Destination MAC	6 B	Unicast, multicast, or broadcast FF:FF:FF:FF:FF:FF
Source MAC	6 B	Sender
Type / Length	2 B	≥ 0x0600: type (0x0800 IPv4, 0x0806 ARP). ≤ 1500: length
Data	46–1500 B	Padded up to 46 B if shorter
FCS	4 B	CRC-32 over the frame

- **Minimum frame 64 B** comes from CSMA/CD (Chapter 6): 512 bits at 10 Mbps = 51.2 µs.
- **MTU = 1500 B** (largest payload). Maximum frame = 14 + 1500 + 4 = 1518 B.
- MAC address: 48 bits, written as six hex bytes. The lowest bit of the first byte = 1 means a group (multicast) address.

7.3 Efficiency of a frame

● LEVEL 2 · EXAM

Case	Formula	Value
Maximum frame, no preamble	1500 / 1518	98.81%
Maximum, with preamble 8 B + gap 12 B	1500 / 1538	97.53%
Minimum frame, no preamble	46 / 64	71.88%
Minimum, with preamble + gap	46 / 84	54.76%

Padding. If the payload is 20 B, the sender pads 26 B so that the data field is 46 B. (The higher layer must know the true length.)

7.4 Hub, switch, router: domains

● LEVEL 2 · EXAM

Device	Collision domains	Broadcast domains	Behaviour
Hub (repeater)	1 for all ports	1	Copies bits to all ports; half-duplex, CSMA/CD
Switch (bridge)	1 per port	1	Forwards frames by MAC; full-duplex possible, no collisions on a switched link
Router	1 per port	1 per interface	Forwards packets by IP; stops broadcasts

Learning switch (short)

- **Learn:** note the source MAC and the incoming port in the MAC table.
- **Forward:** if the destination MAC is in the table, send only to that port. If it is on the *same* port as arrival, drop (filter).
- **Flood:** if the destination is unknown or broadcast, send out of **all ports except the incoming one**.

```
Example:  [A]--p1-[ Switch ]-p2--[B]      A sends to B (table empty)
                  |p3                    1) learn  A -> p1
                 [C]                     2) B unknown -> flood to p2 and p3
```

7.5 Speed variants

● LEVEL 2 · EXAM

Name	Rate	Slot time	Note
10BASE-T	10 Mbps	51.2 µs	Classic CSMA/CD, 512-bit slot
100BASE-TX (Fast Ethernet)	100 Mbps	5.12 µs	Same 64 B minimum, so shorter diameter
1000BASE-T (Gigabit)	1 Gbps	4.096 µs	Half-duplex used carrier extension to 512 B; switches use full-duplex (no CSMA/CD)

QUICK TRICK

With **full-duplex switches** there are no collisions, so CSMA/CD and the minimum-size rule are not needed for correctness (the 64 B minimum is kept for compatibility).

7.6 Common mistakes

COMMON MISTAKE

- Counting preamble in the 64 B / 1518 B frame size. The frame size excludes preamble and SFD.
- Saying a switch separates broadcast domains. It does not (a router does).
- Thinking each port of a hub is its own collision domain (it is one shared domain).
- Forgetting padding when the payload is less than 46 B.

GATE QUICK CHECK — Chapter 7

```
Frame  : 14 header + 46..1500 data + 4 FCS = 64..1518 B     preamble+SFD = 8 B extra
Eff max: 1500/1518 = 98.8%     Eff min: 46/64 = 71.9%       MTU = 1500 B
Domains: hub 1 collision | switch 1 per port | router 1 broadcast per interface
Switch : learn source, forward known, flood unknown/broadcast (not back out of incoming port)
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Minimum and maximum Ethernet frame sizes (excluding preamble)?

Q2. Ethernet broadcast address?

Q3. A payload is 20 bytes. How many padding bytes?

Q4. Which device separates broadcast domains?

● LEVEL 2 · EXAM

Q5. Efficiency of a maximum-size frame, preamble excluded.

Q6. Time to transmit a minimum frame at 100 Mbps.

Q7. A switch with an empty table receives a frame from A on port 1 for host B. What does it do?

Q8. Maximum frame size including preamble and SFD.

● LEVEL 3 · GATE

Q9. A 4-port switch has each port connected to a separate hub with 3 hosts. Count collision domains and broadcast domains.

Q10. Efficiency of a minimum frame when preamble (8 B) and inter-frame gap (12 B) are counted.

Q11. How many Ethernet frames are needed to carry 4000 B of data (ignore higher-layer headers)?

Answers

Q	Solution
1	**64 B** and **1518 B**.
2	**FF:FF:FF:FF:FF:FF**.
3	46 − 20 = **26 bytes**.
4	A **router**.
5	1500/1518 = **98.81%**.
6	512 bits / 10^8 = **5.12 µs**.
7	Learns A → port 1, then **floods** the frame to all ports except port 1.
8	7 + 1 + 14 + 1500 + 4 = **1526 B**.
9	One collision domain per switch port: **4** (each hub plus its hosts is shared). Broadcast domain: **1**.
10	46/(64 + 8 + 12) = 46/84 = **54.76%**.
11	Each frame carries at most 1500 B: 1500 + 1500 + 1000, so **3 frames** (⌈4000/1500⌉).

Next: Chapter 8 — Routing: Distance Vector and Link State.